ShopEZ – E-Commerce Backend API

Name:Neela Lavnaya

---

1. □ Project Overview

ShopEZ is a backend REST API developed using **ASP.NET Core Web API** and **Entity Framework Core**.
This project enables dynamic product management, user authentication, and order processing for an e-commerce system.

The backend replaces static frontend data with a **real SQL Server database** and provides scalable APIs for full-stack integration.

---

2. □ Project Objectives

- Implement product management system (CRUD)
- Build REST APIs for frontend integration
- Process customer orders dynamically
- Store data using SQL Server
- Apply business logic and validations
- Implement secure authentication using JWT

---

3. □ Technology Stack

Layer	Technology
Backend	ASP.NET Core Web API
ORM	Entity Framework Core
Database	SQL Server
Authentication	JWT Authentication
Tools	Visual Studio, Postman

---

4. □ Architecture

The project follows **Layered Architecture**:

```
Controller → Service → Repository → DbContext
```

Responsibilities

- **Controller** → Handles HTTP requests and responses
- **Service** → Business logic and validations
- **Repository** → Database operations
- **DbContext** → Entity Framework database interaction

---

5. □ Project Structure

```
BackendProject
│
├── Controllers
├── Models
├── DTOs
├── Data
├── Repositories
├── Services
└── Program.cs
```

---

6. □ Database Design

Users Table

- UserId (int)
- Name (varchar)
- Email (varchar)
- Password (varchar)
- Role (varchar)

Products Table

- ProductId (int)
- Name (varchar)
- Description (varchar)
- Price (decimal)
- ImageUrl (varchar)
- Stock (int)

Orders Table

- OrderId (int)
- UserId (int)
- OrderDate (datetime)
- TotalAmount (decimal)

OrderItems Table

- OrderItemId (int)
- OrderId (int)
- ProductId (int)
- Quantity (int)
- Price (decimal)

---

7. □ Relationships

- One User → Many Orders
- One Order → Many OrderItems
- One Product → Many OrderItems

---

8. □ Authentication System

Register API

```
POST /api/authenticate/register
```

Login API

```
POST /api/authenticate/login
```

JWT Token Usage

```
Authorization: Bearer <token>
```

---

9. □ Product APIs

Method	Endpoint	Description
GET	/api/products	Get all products
GET	/api/products/{id}	Get product by ID
POST	/api/products	Add product (Admin)
PUT	/api/products/{id}	Update product
DELETE	/api/products/{id}	Delete product

---

10. □ Order APIs

Method	Endpoint	Description
POST	/api/orders	Create order
GET	/api/orders	Get all orders

Method	Endpoint	Description
GET	/api/orders/{id}	Get order by ID

---

11. □ Order Processing Logic

When a user places an order:

1. Validate request data
2. Check product existence
3. Validate quantity (> 0)
4. Check stock availability
5. Deduct product stock
6. Calculate total using LINQ
7. Create Order and OrderItems
8. Save to database

---

12. □ Asynchronous Programming

All database operations use async/await:

```
await _context.SaveChangesAsync();
```

---

13. □ Dependency Injection

The project uses built-in DI:

- DbContext injected into repositories
- Services injected into controllers
- Loose coupling maintained

---

14. □ Validation & Exception Handling

- Input validation using Data Annotations
- Null checks implemented
- Product existence validation
- Quantity validation
- Proper HTTP status codes returned

---

15. □ API Testing (Postman)

Steps:

1. Register User
2. Login → Copy JWT Token
3. Add token in headers
4. Test APIs

Header Example:

```
Authorization: Bearer YOUR_TOKEN
```

---

16. □ Sample Request & Response

Create Order Request

```
{
  "items": [
    { "productId": 1, "quantity": 2 }
  ]
}
```

Response

```
{
  "message": "Order created successfully",
  "orderId": 1,
  "totalAmount": 150000
}
```

---

17. □ Swagger Documentation

Run the project and open:

```
https://localhost:<port>/swagger
```

---

18. ▶□ How to Run the Project

Step 1: Open Project in Visual Studio

Step 2: Configure Database

Update connection string in `appsettings.json`

Step 3: Run Migration

```
Add-Migration InitialCreate
Update-Database
```

Step 4: Run Project

Press **F5**

Step 5: Test APIs using Swagger/Postman

---

19. □ JWT Configuration

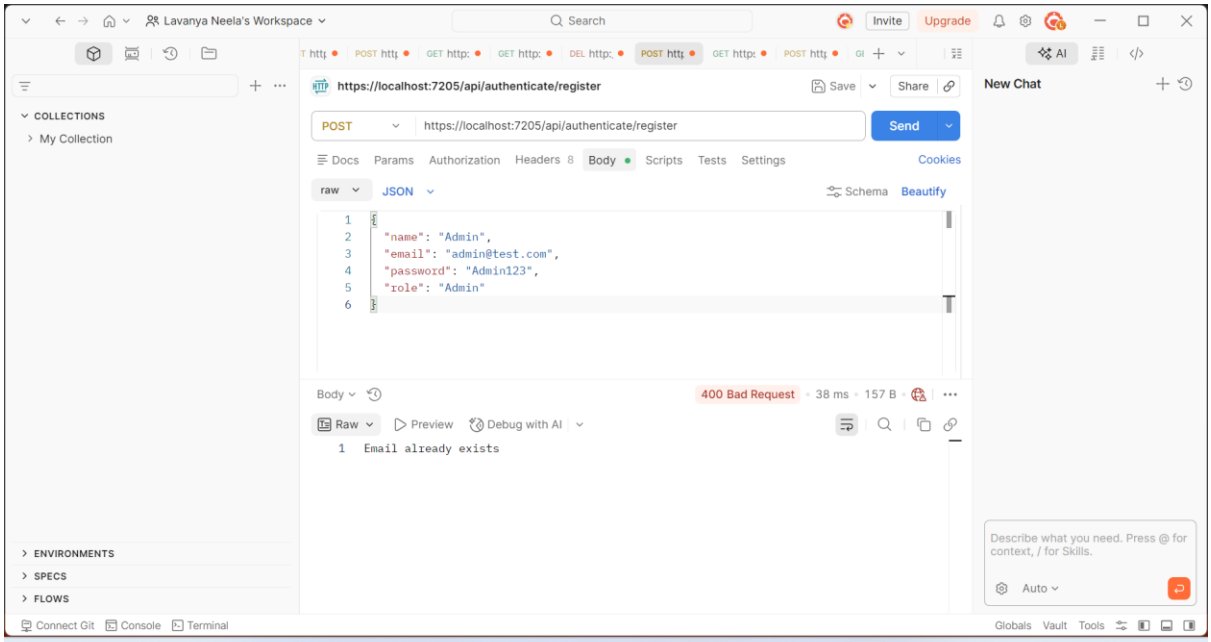
```
"Jwt": {
  "Key": "ThisIsMySuperSecretKey1234567890@123",
  "Issuer": "ShopEZ",
  "Audience": "ShopEZUsers"
}
```

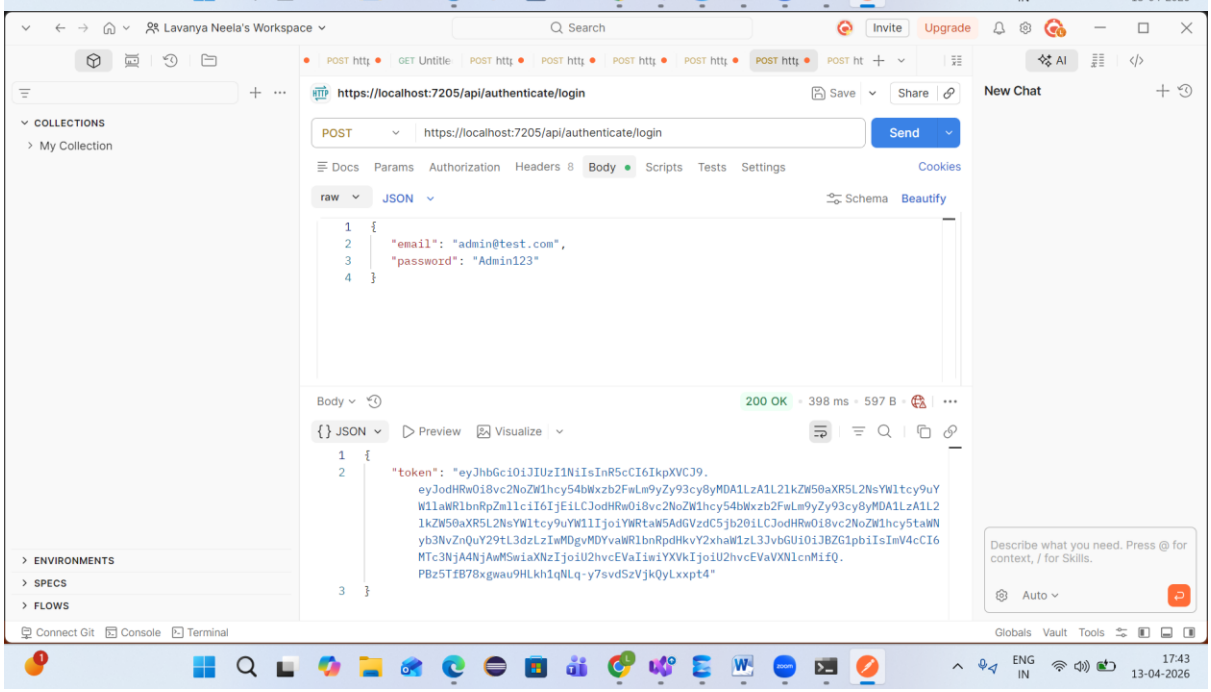
---

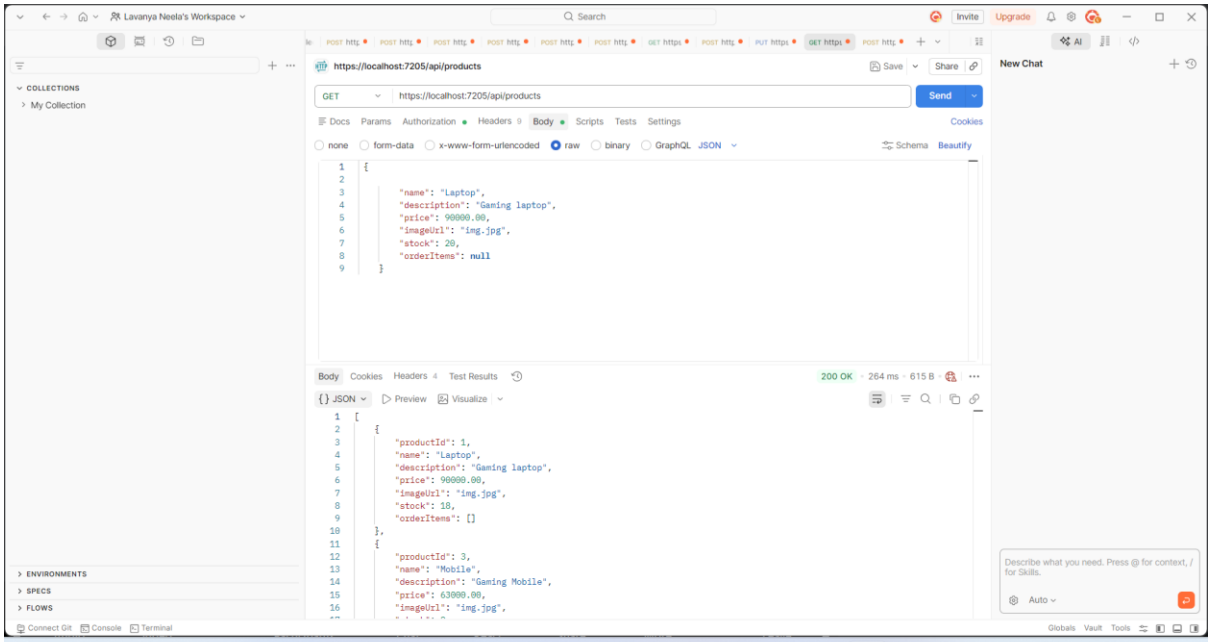
20. □ Features Implemented

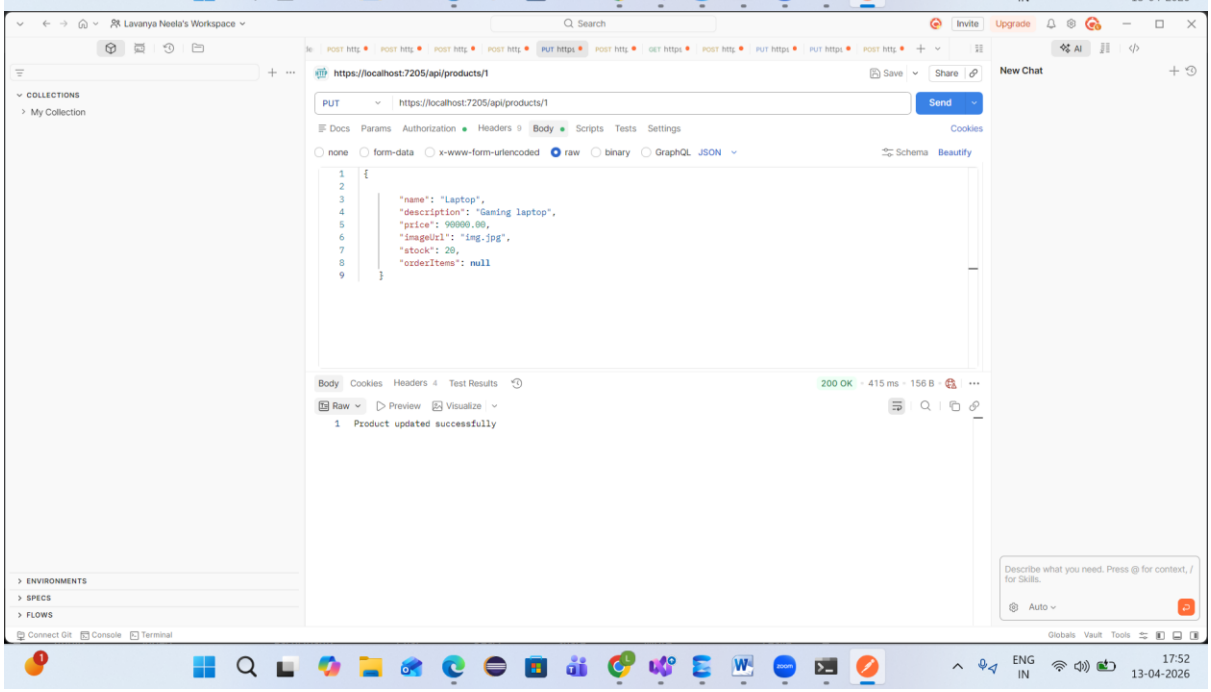
- Product CRUD APIs
- Order Management APIs
- JWT Authentication
- Role-Based Authorization
- Database Integration
- Clean Architecture
- Async Programming

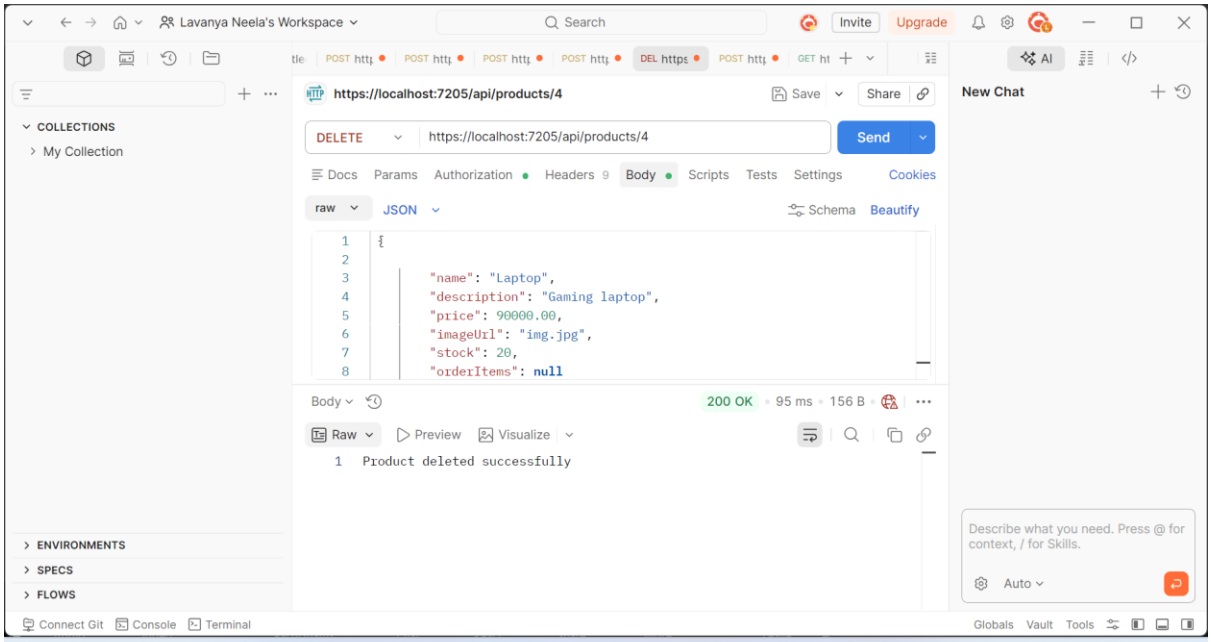
Registration

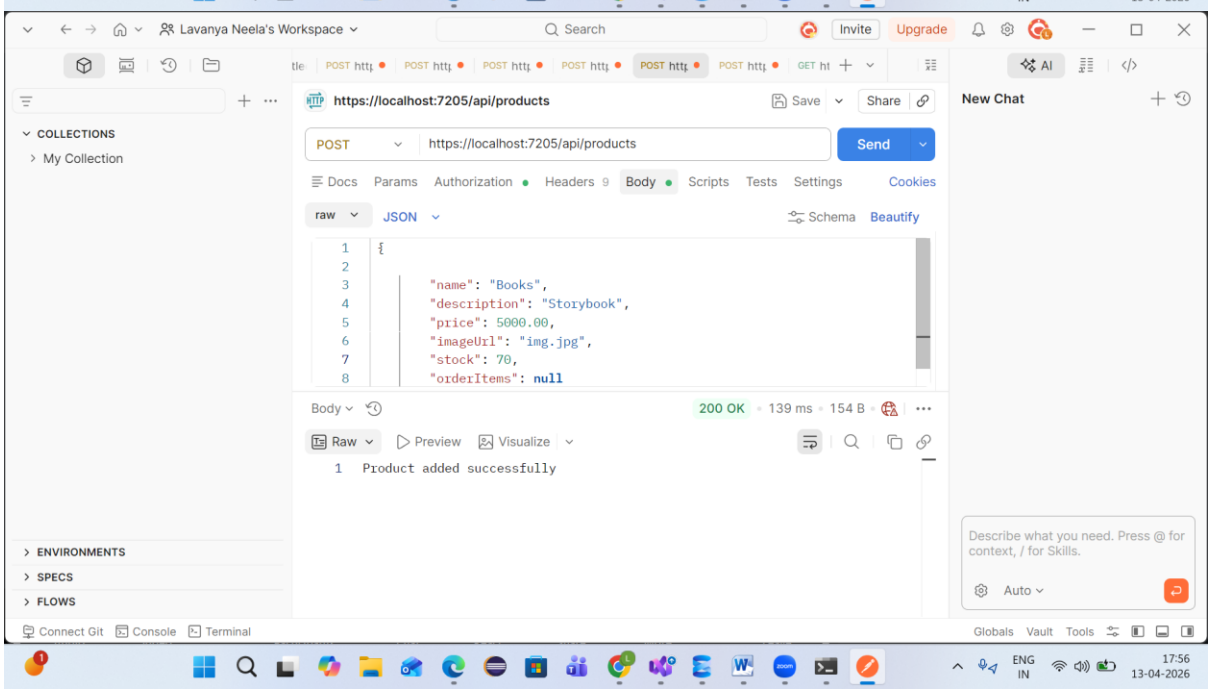

Lavanya Neela's Workspace
Search
Invite
Upgrade
https://localhost:7205/api/authenticate/register
Save
Share
New Chat
COLLECTIONS
My Collection
POST
https://localhost:7205/api/authenticate/register
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Tests
Settings
Cookies
raw
JSON
Schema
Beautify
{
"name": "Admin",
"email": "admin@test.com",
"password": "Admin123",
"role": "Admin"
}
Body
400 Bad Request
38 ms
157 B
Raw
Preview
Debug with AI
Email already exists
Describe what you need. Press @ for context, / for Skills.
Auto
ENVIRONMENTS
SPECS
FLOWS
Connect Git
Console
Terminal
Globals
Vault
Tools
ENG
IN
17:42
13-04-2026

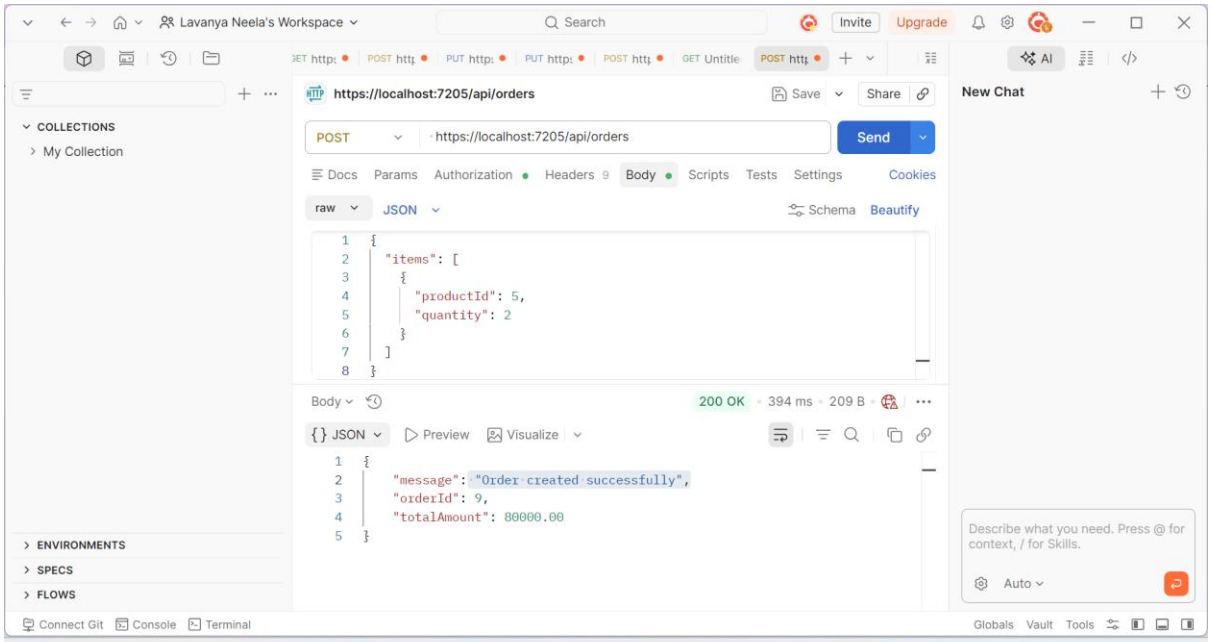

Lavanya Neela's Workspace
Search
Invite
Upgrade
https://localhost:7205/api/authenticate/login
Save
Share
New Chat
COLLECTIONS
My Collection
POST
https://localhost:7205/api/authenticate/login
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Tests
Settings
Cookies
raw
JSON
Schema
Beautify
{
"email": "admin@test.com",
"password": "Admin123"
}
Body
200 OK
398 ms
597 B
JSON
Preview
Visualize
{
"token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
eyJodHRwOi8vc2NoZW1hcy54bWxzb2FwLm9yZy93cy8yMDA1LzA1L2lkZW50aXR5L2NsYW1tcy9uY
W1laWRlbnRpZmllciI6IjEiLCJodHRwOi8vc2NoZW1hcy54bWxzb2FwLm9yZy93cy8yMDA1LzA1L2
1kZW50aXR5L2NsYW1tcy9uYW1lIjoiYWRtaW5AdGVzdC5jb20iLCJodHRwOi8vc2NoZW1hcy5taWN
yb3NvZnQuY29tL3dzLzIwMDgvMDYvaWRlbnRpdHkvY2xhaW1zL3JvbGUiOiJBZG1pbiIsImV4cCI6
MTc3NjA4NjAwMSwiaXNzIjoiU2hvcEVaIiwiYXVkIjoiU2hvcEVaVXNlcnMifQ.
PBz5TfB78xgwau9HLkh1qNLq-y7svdSzVjkQyLxxpt4"
}
Describe what you need. Press @ for context, / for Skills.
Auto
ENVIRONMENTS
SPECS
FLOWS
Connect Git
Console
Terminal
Globals
Vault
Tools
ENG
IN
17:43
13-04-2026

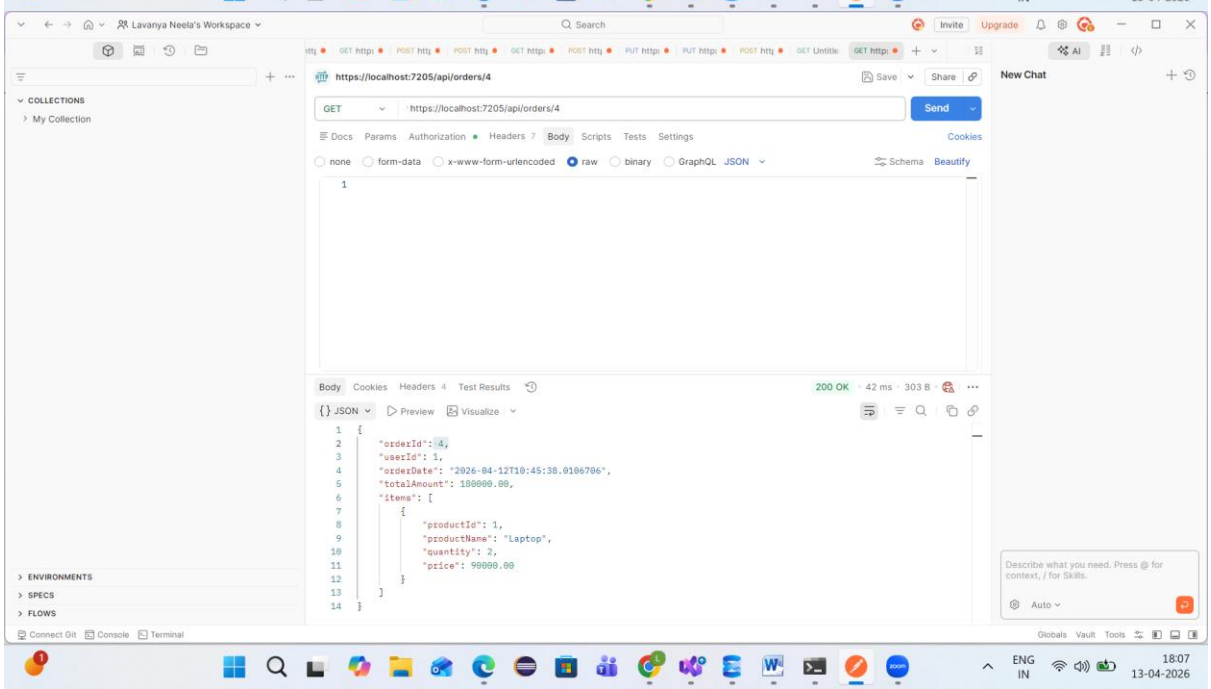

Lavanya Neela's Workspace
Search
Invite
Upgrade
COLLECTIONS
My Collection
https://localhost:7205/api/products
Save
Share
GET
https://localhost:7205/api/products
Send
Docs Params Authorization Headers 9 Body Scripts Tests Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
{
"name": "Laptop",
"description": "Gaming laptop",
"price": 90000.00,
"imageUrl": "img.jpg",
"stock": 20,
"orderItems": null
}
Body Cookies Headers 4 Test Results
200 OK · 264 ms · 615 B
JSON Preview Visualize
[
{
"productId": 1,
"name": "Laptop",
"description": "Gaming laptop",
"price": 90000.00,
"imageUrl": "img.jpg",
"stock": 18,
"orderItems": []
},
{
"productId": 3,
"name": "Mobile",
"description": "Gaming Mobile",
"price": 63000.00,
"imageUrl": "img.jpg",
New Chat
Describe what you need. Press @ for context, / for Skills.
Auto
ENVIRONMENTS
SPECS
FLOWS
Connect Git Console Terminal
Globals Vault Tools
ENG IN
17:45
13-04-2026

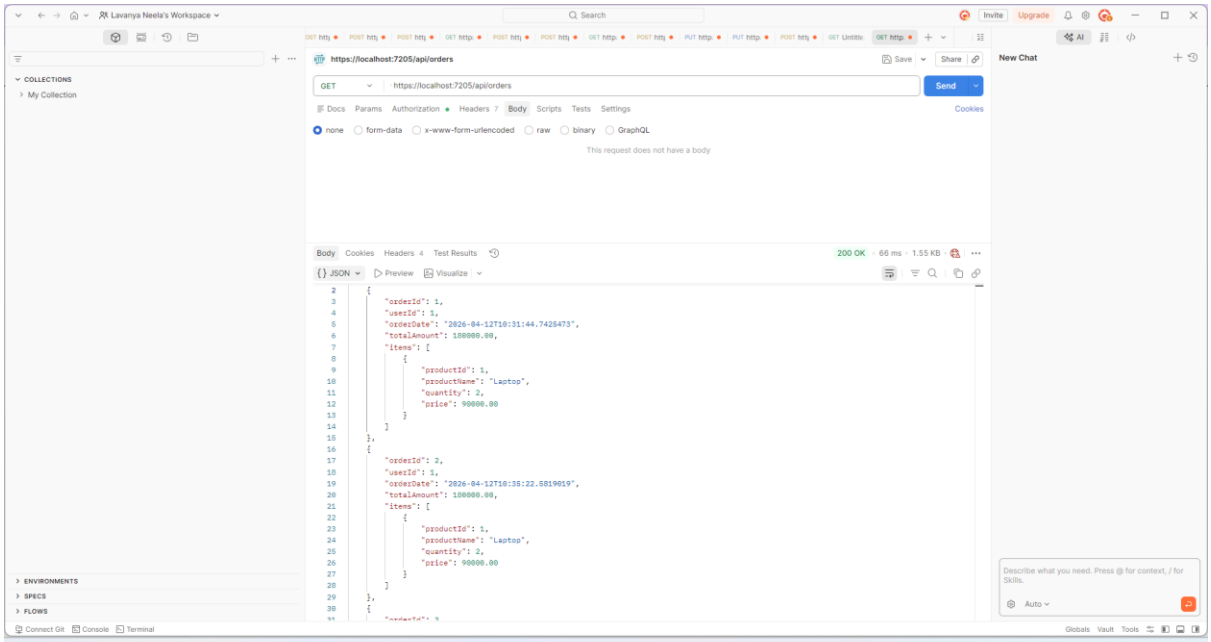

Lavanya Neela's Workspace
Search
Invite
Upgrade
COLLECTIONS
My Collection
https://localhost:7205/api/products/1
Save
Share
PUT
https://localhost:7205/api/products/1
Send
Docs Params Authorization Headers 9 Body Scripts Tests Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
{
"name": "Laptop",
"description": "Gaming laptop",
"price": 90000.00,
"imageUrl": "img.jpg",
"stock": 20,
"orderItems": null
}
Body Cookies Headers 4 Test Results
200 OK · 415 ms · 156 B
Raw Preview Visualize
Product updated successfully
New Chat
Describe what you need. Press @ for context, / for Skills.
Auto
ENVIRONMENTS
SPECS
FLOWS
Connect Git Console Terminal
Globals Vault Tools
ENG IN
17:52
13-04-2026

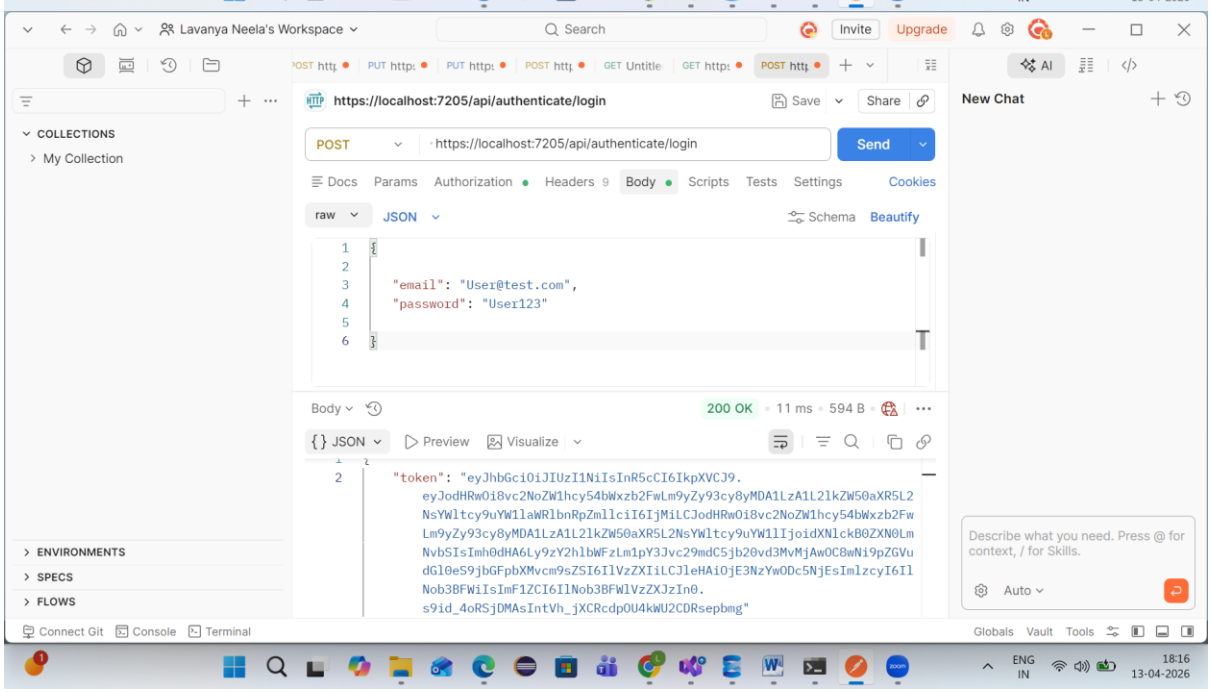

Lavanya Neela's Workspace
Search
Invite
Upgrade
https://localhost:7205/api/products/4
Save
Share
New Chat
DELETE
https://localhost:7205/api/products/4
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Tests
Settings
Cookies
raw
JSON
Schema
Beautify
{
"name": "Laptop",
"description": "Gaming laptop",
"price": 90000.00,
"imageUrl": "img.jpg",
"stock": 20,
"orderItems": null
Body
200 OK
95 ms
156 B
Raw
Preview
Visualize
Product deleted successfully
COLLECTIONS
My Collection
ENVIRONMENTS
SPECS
FLOWS
Describe what you need. Press @ for context, / for Skills.
Auto
Connect Git
Console
Terminal
Globals
Vault
Tools
ENG
IN
17:52
13-04-2026


Lavanya Neela's Workspace
Search
Invite
Upgrade
https://localhost:7205/api/products
Save
Share
New Chat
POST
https://localhost:7205/api/products
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Tests
Settings
Cookies
raw
JSON
Schema
Beautify
{
"name": "Books",
"description": "Storybook",
"price": 5000.00,
"imageUrl": "img.jpg",
"stock": 70,
"orderItems": null
Body
200 OK
139 ms
154 B
Raw
Preview
Visualize
Product added successfully
COLLECTIONS
My Collection
ENVIRONMENTS
SPECS
FLOWS
Describe what you need. Press @ for context, / for Skills.
Auto
Connect Git
Console
Terminal
Globals
Vault
Tools
ENG
IN
17:56
13-04-2026

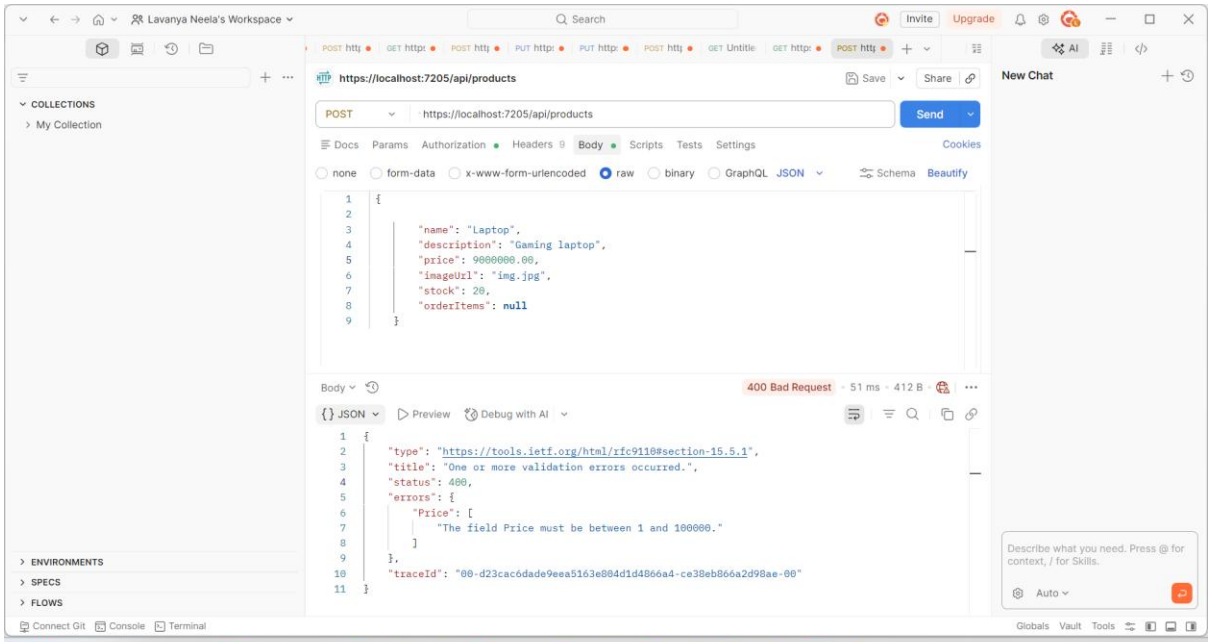

Lavanya Neela's Workspace
Search
Invite
Upgrade
https://localhost:7205/api/orders
Save
Share
New Chat
COLLECTIONS
My Collection
POST
https://localhost:7205/api/orders
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Tests
Settings
Cookies
raw
JSON
Schema
Beautify
{
"items": [
{
"productId": 5,
"quantity": 2
}
]
}
Body
200 OK
394 ms
209 B
JSON
Preview
Visualize
{
"message": "Order created successfully",
"orderId": 9,
"totalAmount": 80000.00
}
ENVIRONMENTS
SPECS
FLOWS
Describe what you need. Press @ for context, / for Skills.
Auto
Connect Git
Console
Terminal
Globals
Vault
Tools
ENG
IN
18:06
13-04-2026

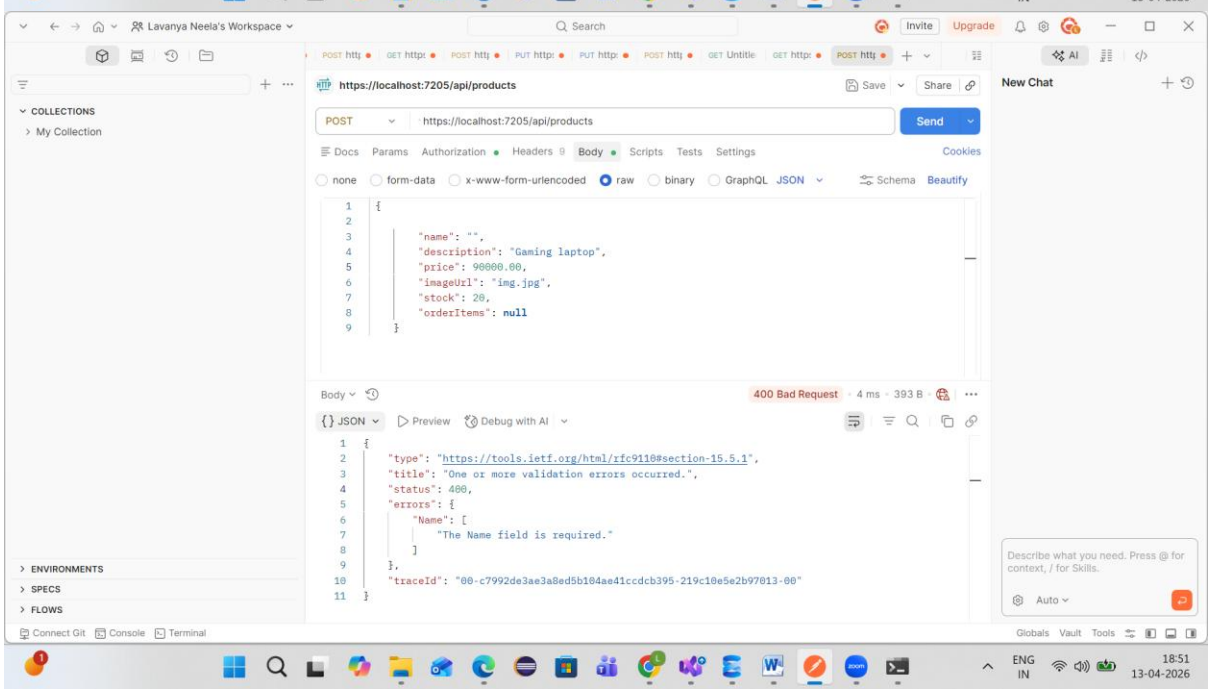

Lavanya Neela's Workspace
Search
Invite
Upgrade
https://localhost:7205/api/orders/4
Save
Share
New Chat
COLLECTIONS
My Collection
GET
https://localhost:7205/api/orders/4
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
Body
Cookies
Headers 4
Test Results
200 OK
42 ms
303 B
JSON
Preview
Visualize
{
"orderId": 4,
"userId": 1,
"orderDate": "2026-04-12T10:45:38.0106706",
"totalAmount": 180000.00,
"items": [
{
"productId": 1,
"productName": "Laptop",
"quantity": 2,
"price": 90000.00
}
]
}
ENVIRONMENTS
SPECS
FLOWS
Describe what you need. Press @ for context, / for Skills.
Auto
Connect Git
Console
Terminal
Globals
Vault
Tools
ENG
IN
18:07
13-04-2026


Lavanya Neela's Workspace
Search
Invite
Upgrade
COLLECTIONS
My Collection
https://localhost:7205/api/orders
GET
https://localhost:7205/api/orders
Send
Docs Params Authorization Headers 7 Body Scripts Tests Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL
This request does not have a body
Body Cookies Headers 4 Test Results
200 OK 66 ms 1.55 KB
JSON Preview Visualize
"orderId": 1,
"userId": 1,
"orderDate": "2026-04-12T10:31:44.7426473",
"totalAmount": 100000.00,
"items": [
"productId": 1,
"productName": "Laptop",
"quantity": 2,
"price": 90000.00
"orderId": 2,
"userId": 1,
"orderDate": "2026-04-12T10:35:22.5819019",
"totalAmount": 100000.00,
"items": [
"productId": 1,
"productName": "Laptop",
"quantity": 2,
"price": 90000.00
ENVIRONMENTS
SPECS
FLOWS
New Chat
Connect Git Console Terminal
Globals Vault Tools
ENG IN
18:08
13-04-2026


Lavanya Neela's Workspace
Search
Invite
Upgrade
COLLECTIONS
My Collection
https://localhost:7205/api/authenticate/login
Save
Share
POST
https://localhost:7205/api/authenticate/login
Send
Docs Params Authorization Headers 9 Body Scripts Tests Settings
Cookies
raw JSON
Schema Beautify
{
"email": "User@test.com",
"password": "User123"
}
Body
200 OK 11 ms 594 B
JSON Preview Visualize
"token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
eyJodHRwOi8vc2NoZW1hcy54bWxzb2FwLm9yZy93cy8yMDA1LzA1L2lkZW50aXR5L2
NsYWltcy9uYW1laWRlbnRpZmllciI6IjMiLCJodHRwOi8vc2NoZW1hcy54bWxzb2Fw
Lm9yZy93cy8yMDA1LzA1L2lkZW50aXR5L2NsYWltcy9uYW1lIjoidXNlckB0ZXN0Lm
NvbSIsImh0dHA6Ly9zY2hlbWFzLm1pY3Jvc29mdC5jb20vd3MvMjAwOC8wNi9pZGVu
dGl0eS9jbGFpbXMvcm9sZSI6IlVzZXIiLCJleHAiOjE3NzYwODc5NjEsImlzcyI6Il
Nob3BFWiIsImF1ZCI6IlNob3BFWlVzZXJzIn0.
s9id_4oRSjDMAsIntVh_jXCRcdpOU4kWU2CDRsepbmg"
New Chat
Describe what you need. Press @ for context, / for Skills.
Auto
ENVIRONMENTS
SPECS
FLOWS
Connect Git Console Terminal
Globals Vault Tools
ENG IN
18:16
13-04-2026

Error Cases





21. ⍰ Conclusion

The ShopEZ backend system successfully provides a scalable and secure API for e-commerce operations.
It follows best practices such as layered architecture, dependency injection, and proper validation.

This backend is fully ready for integration with frontend applications.